

Supplementary Material

Route or Distill? Investigating Lightweight Perceptual Skill Routing for Visual Quadruped Locomotion

I. SKILL TRAINING DETAILS

Both integration approaches use the same library of four visual locomotion policies: baseline running, stair traversal, pit climbing, and gap jumping. We first train the baseline on slopes and rough terrain, then fine-tune separate copies on each obstacle family. The resulting policies are frozen before routing or distillation.

A. Architecture and Depth Observations

Each policy combines a depth CNN, a proprioceptive context encoder, and an asymmetric actor-critic [1]. The CNN processes 48×64 depth images using two eight-channel convolutions with kernel sizes 5×5 and 3×3 , unit stride, and 2×2 max pooling after the first convolution, followed by fully connected layers of widths (128, 64). The context encoder processes 20 consecutive 45-dimensional observations containing commands, projected gravity, angular velocity, joint positions and velocities, and previous actions. Its MLP has widths (256, 128, 80) and outputs Gaussian parameters for a 16-dimensional latent context and 24 explicit estimates: body linear velocity, 17 link-contact states, and four terrain-relative foot heights. The actor combines the current observation, context, explicit estimates, and visual embedding, using hidden widths (512, 256, 128) to produce 12 joint-position offsets. The training-only critic uses five stacked privileged observations (945 inputs) and hidden widths (1024, 256, 128); a decoder with hidden widths (256, 128) reconstructs the next observation from the context and explicit estimates. Hidden layers use ELU activations.

Depth observations refresh asynchronously at 10 Hz, while the context encoder and actor operate at 50 Hz; the training implementation reevaluates the CNN on the latest held image at each policy step. The critic is evaluated during rollouts, whereas the decoder is used only during training updates. Joint targets $\mathbf{q}^{\text{des}} = \mathbf{q}^{\text{stand}} + 0.25\mathbf{a}$ are tracked at 200 Hz with PD gains $K_p = 30$ and $K_d = 0.75$. Table III summarizes the depth settings; images are cropped, resized, clipped, and normalized to $[0, 1]$. The current configuration uses depth-image noise and randomization settings adapted from [2] and summarized in Table V.

B. Training Terrains

The baseline uses equal proportions of sloped and randomly rough terrain; separate copies are fine-tuned on ascending/descending stairs in equal proportions, recessed pits, or gaps. Training uses a 10×10 terrain grid with 8×8 m tiles, a 4 m central platform, and horizontal/vertical resolutions of 0.1/0.005 m. For difficulty $d \in \{0, 0.1, \dots, 0.9\}$, slopes have gradient $\pm 0.4d$, stair risers have magnitude $0.05 + 0.25d$ m with 0.4 m treads, pits have depth $0.1 + 0.6d$ m, and gaps have width $0.1 + 0.9d$ m, before grid quantization. Implemented rough-terrain heights span $[-0.12, 0.05]$ m. The terrain curriculum increases difficulty after traversing more than 4 m and decreases it when progress is insufficient relative to the commanded distance. A command curriculum expands the baseline forward-velocity range from $[-0.5, 0.5]$ to $[-1, 1]$ m/s and specialist ranges from $[0.5, 1]$ to $[0.5, 1.5]$ m/s when normalized tracking reward exceeds 80% of its maximum.

C. Domain Randomization

Skill training randomizes contact, inertial, and actuator properties using the ranges in Table IV. The baseline additionally receives horizontal velocity perturbations with components bounded by ± 1 m/s every 10 s; these perturbations are disabled during specialist fine-tuning. Proprioceptive observation noise is enabled, while camera-pose, depth-image, and latency randomization follow Table V.

D. Optimization and Reward Settings

Policies are trained in 3,000 parallel Isaac Gym environments with a 200 Hz simulation step, 50 Hz policy updates, and a 20 s episode limit. PPO jointly optimizes the actor, critic, and visual encoder. A separate Adam optimizer trains the context encoder and decoder using explicit-state estimation, next-observation reconstruction, and latent KL regularization, with weights 1, 1, and 2, respectively. The shared PPO optimization settings are listed in Table VII. The baseline running skill is trained first, and separate copies are fine-tuned for stairs, pits, and gaps. Fine-tuning updates the complete network and restores the baseline PPO optimizer state and iteration counter. Table VI reports the configured iteration budgets and skill-dependent entropy coefficients.

Locomotion rewards. Table VIII lists the locomotion reward weights, with stairs and gaps sharing the same settings, with all rewards being largely adapted from [1], [2]. Compared with the baseline, specialists emphasize velocity tracking

TABLE III

DEPTH-CAMERA SETTINGS FOR THE SKILL POLICIES.

Setting	Value
Rendered / network resolution	$120 \times 160 / 48 \times 64$
Crop: top, bottom, left, right (pixels)	12, 0, 7, 9
Refresh rate / input history	10 Hz / 1 image
Horizontal field of view	88°
Camera position (x, y, z) (m)	(0.32, 0, 0.10)
Camera orientation (roll, pitch, yaw) (rad)	(0, 1.71, 0)
Render near / far planes (m)	0.05 / 4.0
Network depth clipping (m)	[0, 3]
Latency / resampling interval	80–142 ms / 5 s
Near / far additive-noise amplitude (m)	0.02 / 0.08
Near–far noise boundary (m)	1.2

TABLE IV

PHYSICAL DOMAIN RANDOMIZATION SHARED BY ALL SKILLS.

Parameter	Range
Contact friction coefficient	[0.2, 1.7]
Added base mass (kg)	[−1, 2]
Center-of-mass offset, each axis (m)	[−0.03, 0.03]
K_p , K_d -gain factor	[0.8, 1.2]
Motor-strength factor	[0.9, 1.1]
Joint armature	[0, 0.015]
Joint friction	[0, 0.20]
Joint damping	[0.25, 0.30]
Control-delay steps	{0, 1}

TABLE V

CAMERA AND DEPTH-IMAGE RANDOMIZATION DURING TRAINING. $\mathcal{U}(a, b)$ DENOTES UNIFORM SAMPLING. CORRUPTIONS PRECEDE CLIPPING, NORMALIZATION, AND RESIZING; PATCH DIMENSIONS REFER TO RENDERED-IMAGE PIXELS.

Parameter	Distribution / Value
Camera translation offsets (x, y, z) (m)	$\pm(0.01, 0.0025, 0.03)$
Camera Euler offsets, each axis (rad)	$\mathcal{U}(-0.0577, 0.0577)$
Image latency (ms)	$\mathcal{U}(80, 142)$
Latency resampling interval (s)	5
Minimum stereo distance (m)	0.175
Additive noise, $0.175 \leq d \leq 1.2$ m	$\mathcal{U}(0, 0.02)$ m
Additive noise, $d > 1.2$ m	$\mathcal{U}(0, 0.08)$ m
Near-field block seed probability	0.008
Block height (μ, σ) (pixels)	(62, 1.5)
Block width (μ, σ) (pixels)	(3, 0.01)
Block replacement depth candidates (m)	{0, 0.25, 0.5, 1, 3}
Near-field bright-pixel (spark) probability	0.02
Spark / nonspark depth after clipping (m)	3 / 0
Sky-region depth threshold (m)	2
Sky-artifact seed probability	10^{-4}
Sky-patch height and width (μ, σ) (pixels)	(2, 3.2) each
Sky replacement depths (m)	{0.6, 1, 1.2, 1.5, 1.8}

TABLE VI

CONFIGURED TRAINING BUDGETS.

Policy	Iterations	Entropy weight
Baseline	10,000	0.01
Stairs	40,000	0.01
Pit / climb	60,000	0.005
Gap / jump	40,000	0.01

and obstacle negotiation while relaxing penalties on vertical motion and joint power. Velocity-tracking rewards exponential squared-error with scale 0.2; linear tracking additionally weights lateral error twice as strongly as forward error and includes a heading-alignment factor [3]. The body-height target is 0.38 m, measured relative to the landing surface during gap training. Foot clearance targets 0.08 m above terrain, accounting for a 0.022 m foot-link offset, with tracking scale 0.01. Touchdown airtime reward uses a 0.25 s target and is suppressed beyond 0.50 s. Soft joint-position, joint-velocity, and torque limits are 90% of their respective limits; the foot-edge threshold is 0.05 m.

Obstacle progress and completion. Stair, pit, and gap training additionally reward newly achieved obstacle progress at 0.4 reward per meter and successful completion with a one-time bonus of $0.2 + 0.5d$, where d is terrain difficulty. Obstacle tracking activates within 1 m of the first edge. Completion requires the base and all feet to clear the final edge by 0.35 and 0.05 m, respectively, together with 0.10 s of vertical contact force above 10 N from at least two feet. A stationary penalty of -0.10 reward/s applies after a 0.10 s grace period when speed remains below 0.05 m/s within 1 m of the obstacle.

Distillation Training: We train a unified visual student using DAgger-style imitation from the frozen specialist library. The student executes actions during simulation rollouts, while ground-truth terrain routing selects the specialist that supplies the action target at each visited state. Training alternates rollout collection with supervised updates minimizing the sum of absolute differences between student and specialist actions. We perform three training runs with separate random seeds, and each run trains for 10,000 iterations; the specialists remain fixed, and student optimization uses only action imitation without reinforcement-learning fine-tuning. Empirically, we found that training beyond 10,000 iterations yielded no additional performance benefits.

II. CLASSIFIER/ROUTING TRAINING AND HYPERPARAMETER SELECTION

A. Neural Classifier Architectures

The feature-based classifier is an MLP with 54 inputs (49 geometric descriptors, roll, pitch, and three-axis angular velocity), hidden widths (512, 256, 128), and a four-class linear output layer. The raw-depth classifier processes a single 48×64 depth

TABLE VII
SHARED OPTIMIZATION SETTINGS.

Hyperparameter	Value
Rollout Steps per-Environment	24
PPO Epochs / Mini batches	5 / 4
Discount / GAE parameter	0.99 / 0.95
Value Clipping / Value-Loss Weight	0.2 / 1.0
Initial PPO learning rate (Adam)	10^{-3}
Adaptive Learning Rate bounds	$[10^{-5}, 10^{-2}]$
Target KL	0.01
Gradient-Norm Limit	1.0
Initial Action Standard Deviation	1.0
Context/Decoder Learning Rate (Adam)	2×10^{-4}
Auxiliary Updates per-PPO Minibatch	1
Explicit / Reconstruction / KL weights	1 / 1 / 2

TABLE VIII
LOCOMOTION REWARD WEIGHTS. ZERO DENOTES AN INACTIVE TERM.

Reward term	Baseline	Stairs / Gap	Pit
Linear-velocity tracking	1	2	2
Yaw-rate tracking	0.5	0.5	0.5
Vertical linear velocity	-2	-0.1	-0.1
Roll/pitch angular velocity	-0.05	-0.05	-0.05
Orientation deviation	0	-1	-0.1
Body-height error	-1	-1	-1
Absolute joint power	-2×10^{-4}	-2×10^{-5}	-2×10^{-6}
Joint acceleration	-2×10^{-7}	-2×10^{-7}	-2×10^{-8}
Joint-position limits	-2	-2	-2
Default-posture deviation	-0.1	0	0
Hip-position deviation	0	-0.15	0.015
Foot Clearance	0.2	0.7	0.3
Feet Airtime	1	0.6	0.6
Foot Stumble	0	-1	-1
Foot contact near terrain edges	0	-1	-1
Standing foot contacts	0.5	0	0
Standing posture deviation	-0.5	0	0

image using two unpadded convolutions with 8 and 16 channels, kernel sizes 5×5 and 3×3 , and unit stride, with 2×2 max pooling after the first convolution. The flattened visual representation is concatenated with the same five robot-state inputs and passed through a 128-unit fully connected layer and a four-class output layer. Both networks use ELU activations and configurable dropout after hidden fully connected layers; the CNN implementation also applies ELU after its final output layer. Output scores are converted to class probabilities by softmax for instantaneous selection or temporal filtering.

B. Geometric Depth Feature Details

Let $D(u, v)$ denote the processed normalized depth at image column u and row v . The binary mask $V(u, v)$ identifies valid measurements: it is computed before invalid-depth filling using $0.02 < D < 1$ and retained through preprocessing. Depth-valued features are divided by the valid depth range $s = 0.98$.

For any pixelwise quantity X , binary inclusion mask W , and image region $\mathcal{A}$, define

$$\langle X \rangle_{W, \mathcal{A}} = \frac{\sum_{(u, v) \in \mathcal{A}} W(u, v) X(u, v)}{\max\left(1, \sum_{(u, v) \in \mathcal{A}} W(u, v)\right)} \quad (1)$$

Here, $W = V$ would average over all valid pixels, and $W = 1$ includes every pixel. Omitting $\mathcal{A}$ denotes the full image. We use $Q_q(X; W)$ for the q -quantile of values selected by mask W ; empty masked statistics return zero.

To reduce the effect of body orientation, a pixelwise flat-ground reference is fitted by ridge regression using roll ϕ and pitch θ . Its intercept b_0 and orientation coefficients b_ϕ, b_θ define the residual depth

$$R(u, v) = D(u, v) - [b_0(u, v) + b_\phi(u, v)\phi + b_\theta(u, v)\theta] \quad (2)$$

The fixed orientation-input scaling is absorbed into these coefficients.

Terrain Discontinuities. Let G be a 3×3 Gaussian kernel and S_x, S_y the horizontal- and vertical-derivative Sobel kernels [4]. With $*$ denoting convolution, the smoothed residual derivatives and their magnitude are

$$g_x = S_x * (G * R), \quad g_y = S_y * (G * R), \quad m = \sqrt{g_x^2 + g_y^2} + \epsilon \quad (3)$$

where $\epsilon = 10^{-6}$ provides numerical regularization. The derivative g_x responds to vertical image edges, while g_y responds to horizontal edges. Derivative statistics use a mask M that includes only pixels with a fully valid 3×3 neighborhood.

Overall edge strength is described by $\langle m \rangle_{M/s}$ and $Q_{0.9}(m; M)/s$. Horizontal and vertical edge energies, their ratio, and the fraction of strong horizontal-edge pixels are

$$E_h = \langle |g_y| \rangle_{M/s}, \quad E_v = \langle |g_x| \rangle_{M/s}, \quad r_{\text{edge}} = \frac{E_h}{\max(E_v, \epsilon)}, \quad d_h = \langle \mathbf{1}[|g_y| > \eta] \rangle_M \quad (4)$$

where $\mathbf{1}[\cdot]$ is the indicator function and $\eta = 0.007$.

To describe repeated horizontal boundaries, let $e_v = \langle |g_y| \rangle_{M, \text{row}(v)}$ be the mean horizontal-edge strength in row v . For an image with H rows, the repetition and maximum-strength features are

$$r_{\text{rep}} = \frac{1}{H} \sum_{v=1}^H \mathbf{1}[e_v > \eta], \quad e_{\text{max}} = \frac{\max_v e_v}{s} \quad (5)$$

Three additional features average the largest 10% of valid values in $|g_y|$, $\max(g_y, 0)$, and $\max(-g_y, 0)$, respectively, and divide each average by s . The latter two distinguish the directions of depth changes.

Multiscale Surface Roughness. Our roughness descriptors summarize absolute local filter responses, following the general use of filter-response statistics for texture characterization [5]. Let A_k be a $k \times k$ averaging kernel. The local deviation from the averaged residual and its global roughness statistic are

$$\delta_k = |R - A_k * R|, \quad \rho_k = \langle \delta_k \rangle_V / s, \quad k \in \{3, 7\} \quad (6)$$

The two smoothing scales capture residual variation at different spatial resolutions, a principle related to multiscale residual representations [6]. The extracted features include both scales, their ratio $\rho_3 / \max(\rho_7, \epsilon)$, and near-field fine roughness $\langle \delta_3 \rangle_{V, \mathcal{N}} / s$, where $\mathcal{N}$ contains the bottom 35% of image rows.

Additional curvature features use the Laplacian magnitude $L = |\nabla^2(G * R)|$ [7], summarized by $\langle L \rangle_M / s$ and $Q_{0.9}(L; M) / s$. For the row-averaged residual $p_v = \langle R \rangle_{V, \text{row}(v)}$, the mean absolute second difference is

$$\kappa = \frac{1}{(H-2)s} \sum_{v=2}^{H-1} |p_{v+1} - 2p_v + p_{v-1}| \quad (7)$$

These descriptors capture fine irregularities, broader surface variation, and changes in the vertical depth profile.

Depth-Return Statistics and Spatial Layout. Related stereo-vision methods aggregate disparity observations and estimate a ground profile to distinguish road surfaces from obstacles [8]. Here, we instead summarize close, far, and invalid depth returns using regional pixel fractions. Two binary masks identify far-or-invalid measurements and close returns:

$$I_{\text{far}} = \mathbf{1}[(D \geq 0.6) \vee (V = 0)], \quad (8)$$

$$I_{\text{close}} = V \mathbf{1}[(D < 0.15) \vee (R < -0.05)]$$

Their global descriptors include the valid fraction $\langle V \rangle_1$ and far-or-invalid fraction $\langle I_{\text{far}} \rangle_1$. Applying the same Gaussian–Sobel operations to V gives its gradient magnitude m_V ; validity-boundary density is $\langle \mathbf{1}[m_V > 0.1] \rangle_1$.

Depth spread is measured by the interquartile range

$$\text{IQR}_D = \frac{Q_{0.75}(D; V) - Q_{0.25}(D; V)}{s}. \quad (9)$$

Let $\mathcal{F}$ contain the top third of image rows. The near–far depth difference is

$$\Delta_D = \frac{\langle D \rangle_{V, \mathcal{N}} - \langle D \rangle_{V, \mathcal{F}}}{s} \quad (10)$$

Close-return fractions are computed over the central 50% of columns and over their intersection with the upper 40% of rows. Each occupancy fraction divides by all pixels in its region, including invalid pixels.

We divide the processed image into a 2×3 grid and compute four statistics within each of its six regions, preserving coarse spatial layout through regional pooling [9]. Each cell $\mathcal{P}$ contributes four features:

$$\mathbf{f}_{\mathcal{P}} = \begin{bmatrix} \langle R \rangle_{V, \mathcal{P}} / s \\ \langle \delta_3 \rangle_{V, \mathcal{P}} / s \\ \langle m \rangle_{M, \mathcal{P}} / s \\ \langle I_{\text{far}} \rangle_{1, \mathcal{P}} \end{bmatrix} \quad (11)$$

Classifier Input. The 25 global descriptors and 24 grid descriptors are concatenated with five body-state inputs: roll and pitch divided by 0.7, and body angular velocities $(\omega_x, \omega_y, \omega_z)$ divided by 4. This produces a 54-dimensional vector $\mathbf{f}$. Before MLP training, each component is standardized using its training-set mean μ_j and population standard deviation σ_j :

$$z_j = \frac{f_j - \mu_j}{\max(\sigma_j, \epsilon)}, \quad j = 1, \dots, 54 \quad (12)$$

TABLE IX

NEURAL REGULARIZATION SEARCH. BOTH REPRESENTATIONS USE THE SAME CARTESIAN-PRODUCT GRID. FINAL VALUES ARE FIXED BY THE PAPER EVALUATOR.

Parameter	Candidate values
Training dropout	[0, 0.10, 0.15, 0.20]
Weight decay	[10^{-6} , 10^{-5} , 10^{-4} , 10^{-3}]
Final feature MLP	Dropout 0; weight decay 10^{-5}
Final raw-depth CNN	Dropout 0.20; weight decay 10^{-5}

C. Classifier Dataset Collection and Processing

We collect classifier data by executing the frozen specialist library under ground-truth terrain routing in simulation, covering rough ground, gaps, stairs, and climbing obstacles. At the 10 Hz depth-camera rate, we record normalized 48×64 depth images and synchronized body orientation and angular velocity. Terrain labels are assigned from the robot’s position and simulator terrain map, with ascending and descending stairs sharing one class. This collection requires no additional locomotion-policy optimization.

We collect classifier data by executing the frozen specialist library under ground-truth routing on mixed-terrain maps containing rough ground, gaps, ascending and descending stairs, pits, and raised platforms, grouped into four terrain classes. Collection uses three seeds, each with 100 parallel environments and 60,000 control steps per environment at 50 Hz, totaling 18 million simulator transitions and 5,730 recorded completed episodes. Each map contains 10×10 terrain patches, with roughness heights in $[-0.05, 0.05]$ m, gap widths of 0.2–0.8 m, stair riser magnitudes of 0.1–0.3 m, and pit depths/platform heights of 0.2–0.35 m. At 10 Hz, we record normalized 48×64 depth images, synchronized body orientation and angular velocity, and position-based terrain labels, yielding 3.6 million labeled observations. A separate flat-ground calibration capture contributes 200,000 observations from one million additional simulator transitions, bringing the total collection budget to 19 million transitions without further policy optimization.

Recorded observations are compiled into environment-disjoint training, validation, and test partitions, retaining temporal order and sequence boundaries for sequential evaluation. Both classifier approaches use the same observations and labels, with roll, pitch, and angular velocity providing shared robot-state inputs. For the feature-based MLP, depth images are converted into geometric descriptors using a separately fitted flat-ground reference, then concatenated with robot-state features and standardized using training-partition statistics. The raw-depth CNN receives the normalized depth images directly alongside the corresponding robot-state inputs.

D. Hyperparameter Selection for Neural Terrain Routing

The hyperparameter procedure separates classifier training from temporal-selector tuning. For each perception representation, a regularization grid defines the candidate neural classifiers, which are evaluated using deterministic inference. Classifier weights are then frozen, and cached outputs on ordered validation sequences are used to tune EMA and fixed-transition Bayesian selectors independently. Instantaneous selection uses the same classifier outputs without an additional temporal parameter search. Automated selection uses validation metrics; test metrics are reserved for reporting. The final evaluation fixes the classifier and selector settings and repeats classifier training across three seeds.

1) *Stage 1: Neural classifier training:* The geometric-feature MLP has three ELU hidden layers of widths 512, 256, and 128. The raw-depth CNN uses convolutional layers with 8 and 16 channels, respective kernel sizes 5×5 and 3×3 , unit strides, and 2×2 max pooling after the first convolution. Its flattened visual representation is concatenated with roll, pitch, and three-axis angular velocity before a 128-unit fully connected hidden layer. These architectures are fixed throughout the regularization search. Geometric features are standardized using training-set statistics, and the geometric reference model is fitted on a separate calibration dataset.

Table IX lists the regularization grid, comprising 16 candidate training configurations per representation. Models are optimized using cross-entropy loss and Adam with learning rate 10^{-3} and minibatch size 256. Search training permits up to 50 epochs, stops after 10 epochs without a decrease in validation loss, and restores the checkpoint with minimum validation loss. Dropout is applied during training and disabled during validation and deterministic inference. The deterministic candidates are ranked by ordered-validation balanced accuracy, with lower negative log-likelihood and then lower Brier score breaking ties. The final feature-based and raw-depth configurations use dropout probabilities 0 and 0.20, respectively, with weight decay 10^{-5} for both.

2) *Stage 2: EMA selector tuning:* EMA smooths the deterministic network scores $\mathbf{z}_t$ according to

$$\bar{\mathbf{z}}_t = \alpha \mathbf{z}_t + (1 - \alpha) \bar{\mathbf{z}}_{t-1}. \quad (13)$$

The highest-scoring class is accepted after it remains the candidate for h consecutive updates. The grid in Table X contains eight combinations of smoothing coefficient and confirmation patience, evaluated independently for each representation. At

TABLE X
EMA SEARCH SPACE AND FINAL SETTINGS, SHARED ACROSS PERCEPTION REPRESENTATIONS.

Parameter	Candidate values	Final
Smoothing α	[0, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1]	0.60
Patience h (frames)	[1, 2]	1

TABLE XI
SUCCESSIVE SEARCH GRIDS FOR THE RETAINED FIXED BAYESIAN FILTER.

Parameter	Candidate values
Initial τ	[0.25, 0.40, 0.50, 0.75, 1, 1.5]
Initial s	0.70, 0.8, 0.9, 0.98
Refined τ	[1, 1.25, 1.5, 1.75, 2, 2.5]
Refined s	[0.85, 0.90, 0.95, 0.97, 0.98, 0.99, 0.995, 0.998]
Final settings	$\tau = 1, s = 0.90$

sequence initialization, the filter adopts the first score vector and its predicted class. The final evaluation uses $\alpha = 0.6$ and $h = 1$ for both representations, retaining smoothing without an additional confirmation delay.

3) *Stage 3: Fixed Bayesian selector tuning*: For the Bayesian filter, the searched parameters are score temperature τ and self-transition probability s . Temperature controls the concentration of the classifier probabilities, $\mathbf{p}_t = \text{softmax}(\mathbf{z}_t/\tau)$. For C terrain classes, the transition matrix has $T_{ii} = s$ and $T_{ij} = (1 - s)/(C - 1)$ for $i \neq j$. The belief update is

$$\mathbf{b}_t \propto \mathbf{p}_t \odot (\mathbf{T}^\top \mathbf{b}_{t-1}), \quad (14)$$

with a uniform initial belief. The observation matrix is the identity and the evidence exponent is fixed to one.

Table XI separates the initial grid used during the first search round and the refined grids used during the second search round. Temperature and persistence are crossed within each grid; the two searches are successive stages rather than one Cartesian product of their combined ranges. Neural weights remain unchanged throughout these evaluations. The final paper configuration uses $\tau = 1$ and $s = 0.90$ for both representations.

4) *Temporal selection criterion and final evaluation*: Temporal candidates are scored using

$$S = (1 - w)A_{\text{bal}} + wA_{\text{tr}} - \lambda_D D - 0.05F - 0.05N_{\text{absent}}, \quad (15)$$

where A_{bal} is balanced accuracy, A_{tr} is accuracy within five classification frames of a terrain transition, D is mean matched delay, F is the false-transition fraction, and N_{absent} counts classes never predicted. Accuracies and rates are expressed as fractions. The initial search uses $(w, \lambda_D) = (0.35, 0.002)$; the refined search uses $(0.40, 0.003)$. In the refined ranking, score differences below 0.003 are resolved by lower delay, higher transition accuracy, lower false-transition rate, and higher balanced accuracy, in that order.

For final evaluation, the compiler partitions independent environment/terrain groups into nominal 60/20/20 training, validation, and test splits, while preserving episode-reset boundaries. The classifier and selector configurations above are fixed before training three classifier replicas with three seeds. Final training increases the epoch limit to 100, retaining the same optimizer, minibatch size, early-stopping patience, and best-checkpoint restoration. No hyperparameter search is repeated during this final evaluation.

III. EXPERIMENTAL EVALUATION DETAILS

A. Offline Experiments

The feature-based MLP and raw-depth CNN use identical environment-disjoint data partitions and four terrain classes: rough ground, gaps, stairs, and climbing obstacles. Both networks are trained with seeds $\{0, 1, 2\}$, while the data partitions remain fixed. Classifier and temporal-filter hyperparameters are fixed before test evaluation. All reported statistics in the main text are computed separately for each classifier seed and summarized as mean $\pm$ sample standard deviation.

Instantaneous Classification. Each held-out frame is classified independently using the maximum-probability class, without temporal processing. We report accuracy, balanced accuracy (mean class recall), macro-F1 (mean class F1), negative log-likelihood, and multiclass Brier score. Row-normalized confusion matrices report the distribution of predicted classes conditional on each ground-truth class.

Sequential Prediction. All six representation-selector combinations replay the same 720,000 ordered test frames, containing 6,829 ground-truth terrain transitions. The three selectors are instantaneous selection, EMA of classifier scores with $\alpha = 0.60$ and patience $h = 1$, and Bayesian filtering with temperature $\tau = 1$ and self-transition probability $s = 0.90$. The Bayesian transition matrix distributes the remaining probability uniformly among the other three classes and uses a uniform initial

TABLE XII

CLOSED-LOOP EVALUATION TERRAIN PARAMETERS. STAIRS INCLUDE BOTH ASCENDING AND DESCENDING SEGMENTS; TREAD WIDTH IS 0.4 M.

Parameter	Easy	Medium	Hard
Roughness height range (m)	± 0.03	± 0.06	± 0.09
Gap width (m)	0.40	0.60	0.80
Stair riser magnitude (m)	0.15	0.20	0.25
Pit depth (m)	0.25	0.30	0.35

belief. Filter state is reset at each recorded sequence boundary, and adjacent frames from different sequences are excluded from transition statistics.

Transition Metrics. False-transition rate counts predicted label changes when the ground-truth label is unchanged, divided by all within-sequence adjacent-frame opportunities. For each ground-truth transition, delay is the number of classification frames until the first matching prediction, restricted to the target terrain segment. A transition is missed if no matching prediction occurs before the next terrain change or sequence end; mean delay includes matched transitions only, whereas miss rate uses all ground-truth transitions. Response distributions include both matched delays and misses, with zero delay denoting agreement at the transition frame rather than verified anticipatory or persistent selection.

Temporal Localization and Visualization. Transition-window accuracy is computed over the union of windows extending five classification frames before and after each terrain transition, clipped to sequence boundaries; non-transition accuracy uses the remaining frames. Segment duration is measured from each transition to the next terrain change or sequence end, with short segments defined as those containing at most ten frames. Saved per-frame scores, filtered beliefs, selected labels, ground-truth labels, and sequence indices support reproducible response plots and depth-image case studies without repeating classifier evaluation.

B. Online Closed-Loop Locomotion Experiments

Evaluation Protocol. We compare all six representation-selector combinations, ground-truth terrain routing, and a distilled policy using the same frozen specialist library. Each method completes 900 evaluation episodes: 100 per difficulty-seed combination across three difficulty levels and evaluation seeds. Each combination comprises ten tracks evaluated over ten episodes each, using ten parallel environments. The reported experiments use one classifier checkpoint per representation (seed 0), selected by structural-test balanced accuracy, with lower NLL and Brier score used as tie-breakers. Classifier training seeds are therefore distinct from the three simulation evaluation seeds.

Terrain and Control Settings. Each track contains five consecutive 8 m terrain segments: rough ground, a gap, a pit, ascending stairs, and descending stairs, with their order shuffled by evaluation seed and matched across methods. Difficulty parameters are given in Table XII. Policies execute at 50 Hz, while routing updates every five control steps (10 Hz), retaining the selected specialist between updates and resetting temporal state between episodes. All methods receive a fixed forward command of 1 m/s, zero lateral command, and a shared heading-aware yaw-rate controller that steers toward the track center. The episode limit is 20 s; terrain and command curricula, external pushes, and camera-pose randomization are disabled.

Locomotion and Routing Metrics. An episode succeeds if the robot reaches within 0.25 m of the course endpoint or survives until timeout; environment termination or departure from the assigned track is a failure. Forward distance is the maximum longitudinal displacement from the episode’s initial position, capped at the remaining course length. Terrain-label disagreement is the fraction of classification updates at which the selected skill differs from the current-terrain reference, and skill-switch rate is the number of selected-skill changes divided by the number of classification updates. These routing metrics are undefined for the distilled controller, which produces actions without an explicit skill label. Only completed episodes within the prescribed track quotas contribute to aggregate performance and routing statistics.

Transition-Specific Outcomes. Terrain boundaries are identified geometrically from longitudinal position, with an approach region extending 2 m before each boundary. Traversal succeeds when the robot crosses the boundary and reaches the end of the following terrain segment; failures during the approach or within that segment are recorded separately. Timeouts and evaluation cutoffs before resolution are censored, while unencountered boundaries are marked unreached. Transition-pair traversal rates use only resolved successes and failures, excluding censored and unreached events.

Same-Trajectory Replay. Recorded depth images and synchronized robot states are replayed through all six classifier-selector pipelines, holding observations fixed within each rollout source. Filters are reset at episode boundaries; invalid inputs retain the previous selection and are excluded from recognition statistics. We compute frame-weighted balanced accuracy and accuracy within and outside five-frame windows around terrain-label transitions, retaining rollout-source identity when aggregating results. Episode outcomes and selected-skill histories are also retained to measure disagreement and switching separately for successful and failed episodes.

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